Year 9

REVISION PACK

Unit 4

Name: _____

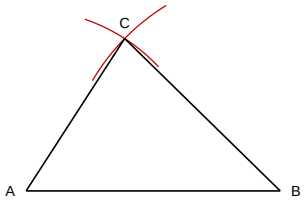
Class: _____

Unit 4 · Constructing triangles (Lesson 4.1)

Worked example: a triangle with three sides

Construct triangle ABC with AB = 7 cm, AC = 5 cm and BC = 6 cm.

Step 1	Draw AB = 7 cm
Step 2	Compasses 5 cm: arc from A
Step 3	Compasses 6 cm: arc from B
Step 4	Join A and B to the crossing



Practice: a triangle with three sides

1.1 Draw AB = 9 cm. Construct triangle ABC with AC = 6 cm and BC = 5 cm.

1.2 Draw AB = 7 cm. Construct triangle ABC with AC = 5 cm and BC = 4 cm.

1.3 Draw AB = 9 cm. Construct triangle ABC with AC = 7 cm and BC = 6 cm.

1.4 Draw AB = 10 cm. Construct triangle ABC with AC = 7 cm and BC = 5 cm.

1.5 Draw AB = 8 cm. Construct triangle ABC with AC = 7 cm and BC = 4 cm.

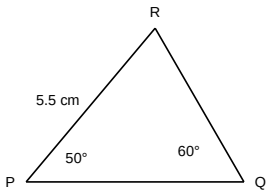
1.6 Draw AB = 6 cm. Construct triangle ABC with AC = 5 cm and BC = 3 cm.

Unit 4 · Constructing triangles (Lesson 4.1)

Worked example: a triangle with a side and two angles

Draw triangle PQR with PQ = 6 cm, angle P = 50° and angle Q = 60°. Measure the length of PR.

Step 1	Draw PQ = 6 cm
Step 2	Measure 50° at P
Step 3	Measure 60° at Q
Step 4	The lines cross at R
Step 5	Answer: PR = 5.5 cm



Practice: a triangle with a side and two angles

1.7 Draw triangle PQR with PQ = 7 cm, angle P = 45° and angle Q = 70°. Measure the length of PR.

1.8 Draw triangle PQR with PQ = 8 cm, angle P = 35° and angle Q = 80°. Measure the length of PR.

1.9 Draw triangle PQR with PQ = 6 cm, angle P = 50° and angle Q = 65°. Measure the length of PR.

1.10 Draw triangle PQR with PQ = 9 cm, angle P = 40° and angle Q = 60°. Measure the length of PR.

1.11 Draw triangle PQR with PQ = 7 cm, angle P = 55° and angle Q = 50°. Measure the length of PR.

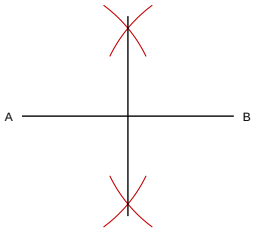
1.12 Draw triangle PQR with PQ = 8 cm, angle P = 30° and angle Q = 70°. Measure the length of PR.

Unit 4 · Bisectors (Lesson 4.2)

Worked example: the perpendicular bisector of a line

Draw a line AB 7 cm long. Construct the perpendicular bisector of AB.

Step 1	Compasses more than half of AB
Step 2	Arcs from A, above and below
Step 3	Same arcs from B
Step 4	Join the two crossings



Practice: the perpendicular bisector of a line

2.1 Draw a line AB 9 cm long. Construct the perpendicular bisector of AB.

2.2 Draw a line CD 8 cm long. Construct the perpendicular bisector of CD.

2.3 Draw a line PQ 10 cm long. Construct the perpendicular bisector of PQ.

2.4 Draw a line XY 6 cm long. Construct the perpendicular bisector of XY.

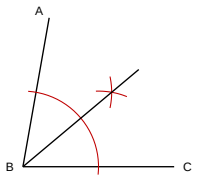
2.5 Draw a line EF 11 cm long. Construct the perpendicular bisector of EF.

2.6 Draw a line MN 5 cm long. Construct the perpendicular bisector of MN.

Worked example: the bisector of an angle

Draw angle ABC = 80°. Construct the bisector of angle ABC.

Step 1	Arc from B on both arms
Step 2	Equal arcs from the two marks
Step 3	Join B to the crossing



Practice: the bisector of an angle

2.7 Draw angle ABC = 70°. Construct the bisector of angle ABC.

2.8 Draw angle PQR = 50°. Construct the bisector of angle PQR.

2.9 Draw angle XYZ = 110°. Construct the bisector of angle XYZ.

2.10 Draw angle DEF = 60°. Construct the bisector of angle DEF.

2.11 Draw angle KLM = 120°. Construct the bisector of angle KLM.

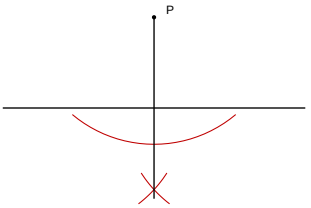
2.12 Draw angle STU = 90°. Construct the bisector of angle STU.

Unit 4 · Perpendiculars and a 30° angle (Lesson 4.2)

Worked example: the perpendicular from a point to a line

Draw a line 10 cm long. Mark a point P 3 cm above the line. Construct the perpendicular from P to the line.

Step 1	Arc from P: cuts the line twice
Step 2	Equal arcs from the two marks, below the line
Step 3	Join P to the crossing



Practice: the perpendicular from a point to a line

3.1 Draw a line 10 cm long. Mark a point P 3.5 cm above the line. Construct the perpendicular from P to the line.

3.2 Draw a line 9 cm long. Mark a point P 3 cm above the line. Construct the perpendicular from P to the line.

3.3 Draw a line 10 cm long. Mark a point P 4 cm above the line. Construct the perpendicular from P to the line.

3.4 Draw a line 8 cm long. Mark a point P 2.5 cm above the line. Construct the perpendicular from P to the line.

3.5 Draw a line 11 cm long. Mark a point P 4.5 cm above the line. Construct the perpendicular from P to the line.

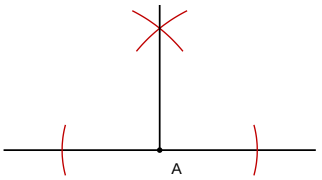
3.6 Draw a line 9 cm long. Mark a point P 2 cm above the line. Construct the perpendicular from P to the line.

Unit 4 · Perpendiculars and a 30° angle (Lesson 4.2)

Worked example: the perpendicular at a point on a line

Draw a line 8 cm long. Mark a point A on the line. Construct the perpendicular to the line at A.

Step 1	Arcs from A on both sides
Step 2	Wider equal arcs from the two marks
Step 3	Join A to the crossing



Practice: the perpendicular at a point on a line

3.7 Draw a line 10 cm long. Mark a point A on the line, 5 cm from the left end. Construct the perpendicular to the line at A.

3.8 Draw a line 9 cm long. Mark a point A on the line, 4 cm from the left end. Construct the perpendicular to the line at A.

3.9 Draw a line 10 cm long. Mark a point A on the line, 6 cm from the left end. Construct the perpendicular to the line at A.

3.10 Draw a line 8 cm long. Mark a point A on the line, 3 cm from the left end. Construct the perpendicular to the line at A.

3.11 Draw a line 11 cm long. Mark a point A on the line, 5 cm from the left end. Construct the perpendicular to the line at A.

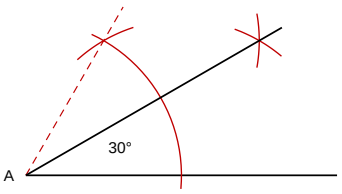
3.12 Draw a line 7 cm long. Mark a point A on the line, 3.5 cm from the left end. Construct the perpendicular to the line at A.

Unit 4 · Perpendiculars and a 30° angle (Lesson 4.2)

Worked example: an angle of 30°

Draw a line 8 cm long from A. Construct an angle of 30° at A.

Step 1	Arc from A: cuts the line
Step 2	Same arc from the mark: 60°
Step 3	Bisect the 60° angle
Step 4	Join: this is 30°



Practice: an angle of 30°

3.13 Draw a line 7 cm long from A. Construct an angle of 30° at A.

3.14 Draw a line 9 cm long from A. Construct an angle of 30° at A.

3.15 Draw a line 10 cm long from A. Construct an angle of 30° at A.

3.16 Draw a line 6 cm long from A. Construct an angle of 30° at A.

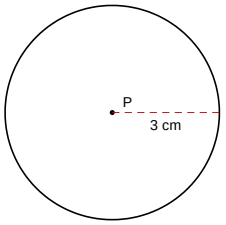
3.17 Draw a line 11 cm long from A. Construct an angle of 30° at A.

3.18 Draw a line 5 cm long from A. Construct an angle of 30° at A.

Unit 4 · Loci and regions (Lesson 4.3)

Worked example: the locus of points from a point

Mark a point P. Draw the locus of all points that are exactly 3 cm from P.

Step 1 Compasses 3 cm	
Step 2 Point on P	
Step 3 Draw a full circle	

Practice: the locus of points from a point

4.1 Mark a point P. Draw the locus of all points that are exactly 3.5 cm from P.

4.2 Mark a point P. Draw the locus of all points that are exactly 2 cm from P.

4.3 Mark a point P. Draw the locus of all points that are exactly 4 cm from P.

4.4 Mark a point P. Draw the locus of all points that are exactly 4.5 cm from P.

4.5 Mark a point P. Draw the locus of all points that are exactly 1.5 cm from P.

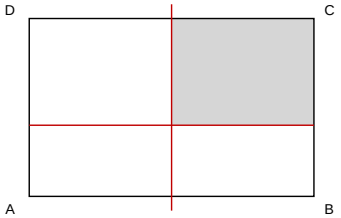
4.6 Mark a point P. Draw the locus of all points that are exactly 5 cm from P.

Unit 4 · Loci and regions (Lesson 4.3)

Worked example: regions in a rectangle

Draw rectangle $ABCD$ with AB = 8 cm and BC = 5 cm. Shade the region inside the rectangle of points that are more than 2 cm from AB **and** closer to C than to D.

Step 1	Line 2 cm above AB
Step 2	Perpendicular bisector of DC
Step 3	Shade above the line, on C's side



Practice: regions in a rectangle

4.7 Draw rectangle $ABCD$ with AB = 9 cm and BC = 6 cm. Shade the region inside the rectangle of points that are more than 2 cm from AB **and** closer to D than to C.

4.8 Draw rectangle $ABCD$ with AB = 8 cm and BC = 5 cm. Shade the region inside the rectangle of points that are more than 3 cm from AB **and** closer to D than to C.

4.9 Draw rectangle $ABCD$ with AB = 10 cm and BC = 6 cm. Shade the region inside the rectangle of points that are more than 4 cm from AB **and** closer to C than to D.

4.10 Draw rectangle $ABCD$ with AB = 9 cm and BC = 5 cm. Shade the region inside the rectangle of points that are more than 1 cm from AB **and** closer to C than to D.

4.11 Draw rectangle $ABCD$ with AB = 10 cm and BC = 7 cm. Shade the region inside the rectangle of points that are more than 3 cm from AB **and** closer to C than to D.

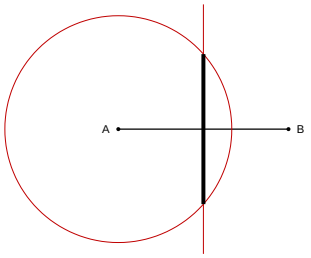
4.12 Draw rectangle $ABCD$ with AB = 7 cm and BC = 5 cm. Shade the region inside the rectangle of points that are more than 2 cm from AB **and** closer to D than to C.

Unit 4 · Loci and regions (Lesson 4.3)

Worked example: a problem with a scale

Towns A and B are 120 km apart. The scale is 1 cm to 20 km. A mast must be less than 80 km from A **and** the same distance from A as from B. Show all the possible positions.

Step 1	Change to cm: $120 \div 20 = 6$ cm, $80 \div 20 = 4$ cm
Step 2	Draw A and B 6 cm apart
Step 3	Circle: radius 4 cm, centre A
Step 4	Perpendicular bisector of AB
Step 5	Answer: the bisector inside the circle



Practice: a problem with a scale

4.13 Towns A and B are 80 km apart. The scale is 1 cm to 10 km. Draw A and B. A phone mast must be less than 60 km from A **and** the same distance from A as from B. Show all the possible positions of the mast.

4.14 Towns A and B are 60 km apart. The scale is 1 cm to 10 km. Draw A and B. A phone mast must be less than 40 km from A **and** the same distance from A as from B. Show all the possible positions of the mast.

4.15 Towns A and B are 140 km apart. The scale is 1 cm to 20 km. Draw A and B. A phone mast must be less than 100 km from A **and** the same distance from A as from B. Show all the possible positions of the mast.

4.16 Towns A and B are 90 km apart. The scale is 1 cm to 10 km. Draw A and B. A phone mast must be less than 70 km from A **and** the same distance from A as from B. Show all the possible positions of the mast.

4.17 Towns A and B are 40 km apart. The scale is 1 cm to 5 km. Draw A and B. A phone mast must be less than 30 km from A **and** the same distance from A as from B. Show all the possible positions of the mast.

4.18 Towns A and B are 100 km apart. The scale is 1 cm to 10 km. Draw A and B. A phone mast must be less than 80 km from A **and** the same distance from A as from B. Show all the possible positions of the mast.

Worked example: map scales

A map has a scale of 1 cm to 4 km.

(a) Two towns are 6.5 cm apart on the map. Work out the real distance.

(b) A road is 30 km long. How long is it on the map?

(a)	Step 1	Map to real: multiply	$6.5 \times 4 = 26$
	Step 2	Answer	26 km
(b)	Step 1	Real to map: divide	$30 \div 4 = 7.5$
	Step 2	Answer	7.5 cm

Practice: map scales

4.19 A map has a scale of 1 cm to 2 km.

(a) Two towns are 8.5 cm apart on the map. Work out the real distance.

(b) A road is 13 km long. How long is it on the map?

4.20 A map has a scale of 1 cm to 25 km.

(a) Two towns are 3.4 cm apart on the map. Work out the real distance.

(b) A road is 105 km long. How long is it on the map?

4.21 A map has a scale of 1 cm to 10 km.

(a) Two towns are 4.7 cm apart on the map. Work out the real distance.

(b) A road is 62 km long. How long is it on the map?

4.22 A map has a scale of 1 cm to 3 km.

(a) Two towns are 9.2 cm apart on the map. Work out the real distance.

(b) A road is 27 km long. How long is it on the map?

4.23 A map has a scale of 1 cm to 20 km.

(a) Two towns are 5.5 cm apart on the map. Work out the real distance.

(b) A road is 130 km long. How long is it on the map?

4.24 A map has a scale of 1 cm to 8 km.

(a) Two towns are 3.5 cm apart on the map. Work out the real distance.

(b) A road is 52 km long. How long is it on the map?

Unit 4 · Can the triangle be constructed? (Lesson 4.1)

Worked example: when a triangle cannot be drawn

Suraya is asked to construct a triangle with sides of length 3 cm, 4 cm and 9 cm. Explain why she cannot do this.

Step 1	Add the two shorter sides	$3 + 4 = 7$
Step 2	Compare with the longest side	$7 < 9$
Step 3	Answer	The two shorter sides are too short to meet, because $3 + 4$ is less than 9

Practice: can the triangle be constructed?

5.1 Can a triangle be constructed with sides of length 5 cm, 6 cm and 8 cm?
Give a reason for your answer.

5.2 Can a triangle be constructed with sides of length 2 cm, 5 cm and 10 cm?
Give a reason for your answer.

5.3 Can a triangle be constructed with sides of length 4 cm, 4 cm and 9 cm?
Give a reason for your answer.

5.4 Can a triangle be constructed with sides of length 6 cm, 7 cm and 12 cm?
Give a reason for your answer.

5.5 Can a triangle be constructed with sides of length 3 cm, 5 cm and 8 cm?
Give a reason for your answer.

5.6 Can a triangle be constructed with sides of length 7 cm, 9 cm and 15 cm?
Give a reason for your answer.